

Alpha decay 3

In this section we derive a regression formula from the data set

	$\log \tau_{1/2}$	E
^{226}U	-1.31	7.70
^{228}U	6.30	6.80
^{230}U	14.4	5.99
^{232}U	21.5	5.41
^{234}U	29.7	4.86
^{236}U	34.2	4.57
^{238}U	39.5	4.27

From the previous sections we have

$$\log \tau_{1/2} \propto \gamma$$

and

$$\gamma = C_1 \frac{Z}{\sqrt{E}} - C_2 \sqrt{Zr_1}$$

Combine the formulas as

$$\log \tau_{1/2} = \beta_1 \frac{90}{\sqrt{E}} + \beta_0$$

The nuclear radius $r_1 \propto A^{1/3}$ is discarded because it does not change very much.

By ordinary least squares we obtain

$$\begin{aligned}\beta_1 &= 3.682 \\ \beta_0 &= -120.8\end{aligned}$$

Hence

$$\log \tau_{1/2} = 3.682 \frac{90}{\sqrt{E}} - 120.8$$

Eigenmath script